# National Testing Agency

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|---|---|
| **Question Paper Name :** | B Tech 4 Apr 2025 Shift 2 |
| **Subject Name :** | B. Tech |
| **Duration :** | 180 |
| **Total Marks :** | 300 |
| **Display Marks:** | Yes |

## B. Tech

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|---|---|
| **Group Number :** | 1 |
| **Group Maximum Duration :** | 0 |
| **Group Minimum Duration :** | 180 |
| **Show Attended Group? :** | No |
| **Edit Attended Group? :** | No |
| **Break time :** | 0 |
| **Group Marks :** | 300 |

## Mathematics Section A

| | |
|---|---|
| **Section Number :** | 1 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 1 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $\mathrm{A} = \{-3, -2, -1, 0, 1, 2, 3\}$ and R be a relation on A defined by $x\mathrm{R}y$ if and only if $2x - y \in \{0, 1\}$. Let $l$ be the number of elements in $R$. Let $m$ and $n$ be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l + \mathrm{m} + \mathrm{n}$ is equal to:

**Options :**

A. 17

B. 18

C. 15

D. 16

**Question Number : 2 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the domains of the functions $f(x) = \log_4 \log_3 \log_7 \left(8 - \log_2 \left(x^2 + 4x + 5\right)\right)$ and $\mathrm{g}(x) = \sin^{-1}\left(\frac{7x+10}{x-2}\right)$ be $(\alpha, \beta)$ and $[\gamma, \delta]$, respectively. Then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is equal to :

**Options :**

A. 15

B. 13

C. 16

D. 14

**Question Number : 3 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the mean and the standard deviation of the observation 2,3,3,4,5,7, $a, b$ be 4 and $\sqrt{2}$ respectively. Then the mean deviation about the mode of these observations is :

**Options :**

A. $\frac{1}{2}$

B. $\frac{3}{4}$

C. 1

D. 2

**Question Number : 4 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the sum of the focal distances of the point P(4,3) on the hyperbola H : $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $8\sqrt{\frac{5}{3}}$. If for H , the length of the latus rectum is $l$ and the product of the focal distances of the point P is m , then $9l^2 + 6$ m is equal to :

**Options :**

A. 187

B. 184

C. 186

D. 185

**Question Number : 5 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let a $> 0$. If the function $f(x) = 6x^3 - 45ax^2 + 108a^2x + 1$ attains its local maximum and minimum values at the points $x_1$ and $x_2$ respectively such that $x_1x_2 = 54$, then a $+ x_1 + x_2$ is equal to :

**Options :**

A. 15

B. 13

C. 24

D. 18

**Question Number : 6 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ satisfy $A^n = A^{n-2} + A^2 - I$ for $n \geqslant 3$. Then the sum of all the elements of $A^{50}$ is :

**Options :**

A. 44

B. 39

C. 52

D. 53

**Question Number : 7 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If a curve $y = y(x)$ passes through the point $\left(1, \frac{\pi}{2}\right)$ and satisfies the differential equation $\left(7x^4 \cot y - e^x \operatorname{cosec} y\right)\frac{dx}{dy} = x^5, x \geq 1$, then at $x = 2$, the value of $\cos y$ is :

**Options :**

A. $\frac{2e^2 + e}{64}$

B. $\frac{2e^2 - e}{64}$

C. $\frac{2e^2 - e}{128}$

D. $\frac{2e^2 + e}{128}$

**Question Number : 8 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If the sum of the first 20 terms of the series $\frac{4 \cdot 1}{4 + 3 \cdot 1^2 + 1^4} + \frac{4 \cdot 2}{4 + 3 \cdot 2^2 + 2^4} + \frac{4 \cdot 3}{4 + 3 \cdot 3^2 + 3^4} + \frac{4 \cdot 4}{4 + 3 \cdot 4^2 + 4^4} + \ \ldots \cdot$ is $\frac{m}{n}$, where m and n are coprime, then m + n is equal to :

**Options :**

A. 423

B. 421

C. 422

D. 420

**Question Number : 9 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let for two distinct values of p the lines $y = x + p$ touch the ellipse $E : \frac{x^2}{4^2} + \frac{y^2}{3^2} = 1$ at the points A and B . Let the line $y = x$ intersect E at the points C and D . Then the area of the quadrilateral *ABCD* is equal to :

**Options :**

A. 48

B. 20

C. 24

D. 36

**Question Number : 10 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The centre of a circle C is at the centre of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$. Let C pass through the foci $F_1$ and $F_2$ of E such that the circle *C* and the ellipse *E* intersect at four points. Let P be one of these four points. If the area of the triangle $PF_1F_2$ is 30 and the length of the major axis of *E* is 17 , then the distance between the foci of *E* is :

**Options :**

A. 12

B. 26

C. 13

D. $\frac{13}{2}$

**Question Number : 11 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A line passing through the point $A(-2,0)$, touches the parabola $P : y^2 = x - 2$ at the point *B* in the first quadrant. The area, of the region bounded by the line *AB*, parabola *P* and the $x$-axis, is :

**Options :**

A. 3

B. $\frac{7}{3}$

C. $\frac{8}{3}$

D. 2

**Question Number : 12 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the product of $\omega_1 = (8+i)\sin\theta + (7+4i)\cos\theta$ and $\omega_2 = (1+8i)\sin\theta + (4+7i)\cos\theta$ be $\alpha + i\beta$, $i = \sqrt{-1}$ . Let p and q be the maximum and the minimum values of $\alpha + \beta$ respectively. Then p + q is equal to :

**Options :**

A. 130

B. 150

C. 160

D. 140

**Question Number : 13 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If $1^2 \cdot \left({}^{15}C_1\right) + 2^2 \cdot \left({}^{15}C_2\right) + 3^2 \cdot \left({}^{15}C_3\right) + \ \ldots\ + 15^2 \cdot \left({}^{15}C_{15}\right) = 2^m \cdot 3^n \cdot 5^k$, where $m, n, k \in \mathbf{N}$, then m + n + k is equal to :

**Options :**

A. 20

B. 19

C. 18

D. 21

**Question Number : 14 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let A be the point of intersection of the lines $L_1 : \frac{x-7}{1} = \frac{y-5}{0} = \frac{z-3}{-1}$ and $L_2 : \frac{x-1}{3} = \frac{y+3}{4} = \frac{z+7}{5}$. Let B and C be the points on the lines $L_1$ and $L_2$ respectively such that $AB = AC = \sqrt{15}$ . Then the square of the area of the triangle $ABC$ is :

**Options :**

A. 63

B. 57

C. 60

D. 54

**Question Number : 15 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The sum of the infinite series $\cot^{-1}\left(\frac{7}{4}\right)+\cot^{-1}\left(\frac{19}{4}\right)+\cot^{-1}\left(\frac{39}{4}\right)+\cot^{-1}\left(\frac{67}{4}\right)+$ ... . is :

**Options :**

A. $\frac{\pi}{2}+\cot^{-1}\left(\frac{1}{2}\right)$

B. $\frac{\pi}{2}-\cot^{-1}\left(\frac{1}{2}\right)$

C. $\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{2}\right)$

D. $\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{2}\right)$

**Question Number : 16 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The axis of a parabola is the line $y=x$ and its vertex and focus are in the first quadrant at distances $\sqrt{2}$ and $2\sqrt{2}$ units from the origin, respectively. If the point $(1,k)$ lies on the parabola, then a possible value of k is :

**Options :**

A. 8

B. 3

C. 9

D. 4

**Question Number : 17 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $f(x)+2f\left(\frac{1}{x}\right)=x^2+5$ and $2g(x)-3g\left(\frac{1}{x}\right)=x, x>0$. If $\alpha=\int_1^2 f(x)\mathrm{d}x$, and $\beta=\int_1^2 g(x)\mathrm{d}x$, then the value of $9\alpha+\beta$ is :

**Options :**

A. 0

B. 10

C. 1

D. 11

**Question Number : 18 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider two sets A and B, each containing three numbers in A.P. Let the sum and the product of the elements of A be 36 and p respectively and the sum and the product of the elements of B be 36 and $q$ respectively. Let d and D be the common differences of AP's in $A$ and $B$ respectively such that $D = d + 3, d > 0$. If $\frac{p+q}{p-q} = \frac{19}{5}$, then p − q is equal to

**Options :**

A. 540

B. 450

C. 600

D. 630

**Question Number : 19 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the values of p , for which the shortest distance between the lines $\frac{x+1}{3} = \frac{y}{4} = \frac{z}{5}$ and $\vec{r} = (p\hat{i} + 2\hat{j} + \hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ is $\frac{1}{\sqrt{6}}$, be a, b, (a < b). Then the length of the latus rectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is :

**Options :**

A. $\frac{3}{2}$

B. 9

C. 18

D. $\frac{2}{3}$

**Question Number : 20 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $f$ be a differentiable function on **R** such that $f(2) = 1, f'(2) = 4$. Let $\lim_{x \to 0} (f(2 + x))^{3/x} = e^{\alpha}$. Then the number of times the curve $y = 4x^3 - 4x^2 - 4(\alpha - 7)x - \alpha$ meets $x$-axis is :

**Options :**

A. 3

B. 1

C. 2

D. 0

## Mathematics Section B

| | |
|---|---|
| **Section Number :** | 2 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |
| **Number of Questions to be attempted :** | 5 |
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 21 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

If $\int \frac{\left(\sqrt{1+x^2}+x\right)^{10}}{\left(\sqrt{1+x^2}-x\right)^9}\,dx = \frac{1}{m}\left(\left(\sqrt{1+x^2}+x\right)^n\left(n\sqrt{1+x^2}-x\right)\right)+C$ where C is the constant of integration and $m, n \in \mathbf{N}$, then $m+n$ is equal to _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 22 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

A card from a pack of 52 cards is lost. From the remaining 51 cards, n cards are drawn and are found to be spades. If the probability of the lost card to be a spade is $\frac{11}{50}$, then n is equal to _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 23 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Let the three sides of a triangle ABC be given by the vectors $2\hat{i}-\hat{j}+\hat{k}, \hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$. Let $G$ be the centroid of the triangle $ABC$. Then $6\left(|\overrightarrow{AG}|^2+|\overrightarrow{BG}|^2+|\overrightarrow{CG}|^2\right)$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 24 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

If $\alpha$ is a root of the equation $x^2+x+1=0$ and $\sum_{k=1}^{n}\left(\alpha^k+\frac{1}{\alpha^k}\right)^2=20$, then n is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 25 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Let m and n, $(m < n)$, be two 2-digit numbers. Then the total numbers of pairs $(m, n)$, such that $\gcd(m, n) = 6$, is _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

## Physics Section A

| | |
|---|---|
| **Section Number :** | 3 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 26 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A finite size object is placed normal to the principal axis at a distance of 30 cm from a convex mirror of focal length 30 cm . A plane mirror is now placed in such a way that the image produced by both the mirrors coincide with each other. The distance between the two mirrors is :

**Options :**

A. 45 cm

B. 15 cm

C. 22.5 cm

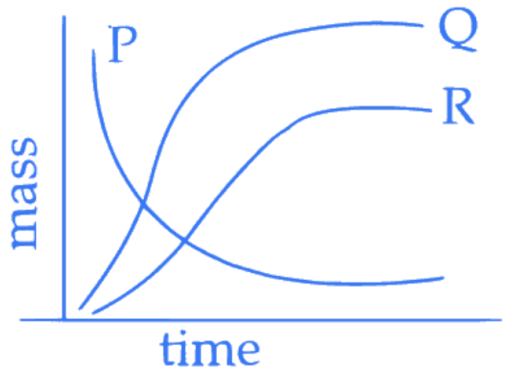
D. 7.5 cm

**Question Number : 27 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

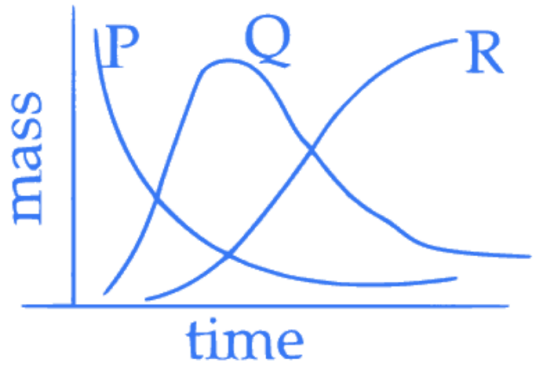
A radioactive material $P$ first decays into $Q$ and then $Q$ decays to non-radioactive material $R$. Which of the following figure represents time dependent mass of $P, Q$ and $R$ ?

**Options :**

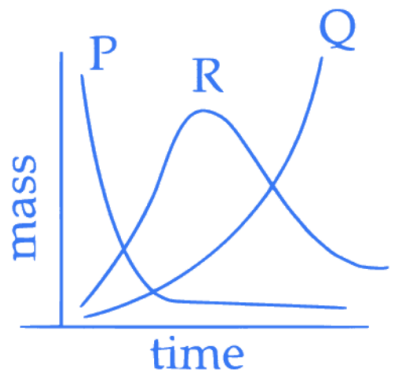
A.



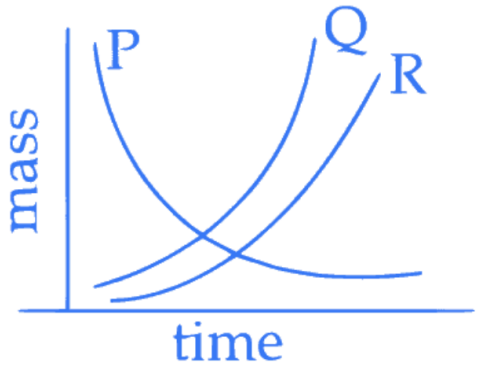

B.




C.




D.



**Question Number : 28 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Displacement of a wave is expressed as $x(t) = 5\cos\left(628t + \frac{\pi}{2}\right)$m. The wavelength of the wave when its velocity is 300 m/s is :

$$(\pi = 3.14)$$

**Options :**

A. 0.33 m

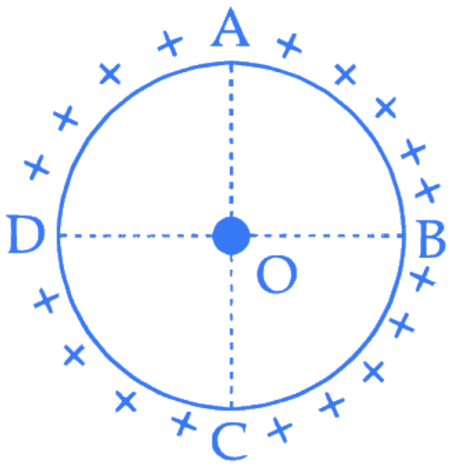
B. 0.5 m

C. 3 m

D. 5 m

**Question Number : 29 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A metallic ring is uniformly charged as shown in figure. AC and BD are two mutually perpendicular diameters. Electric field due to arc $AB$ at ' $O$ ' is ' $E$ ' in magnitude. What would be the magnitude of electric field at ' O ' due to arc ABC ?




**Options :**

A. 2E

B. Zero

C. E/2

D. $\sqrt{2}$ E

**Question Number : 30 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A wheel is rolling on a plane surface. The speed of a particle on the highest point of the rim is 8 m/s. The speed of the particle on the rim of the wheel at the same level as the centre of wheel, will be :

**Options :**

A. $4\sqrt{2}$ m/s

B. 8 m/s

C. 4 m/s

D. $8\sqrt{2}$ m/s

**Question Number : 31 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

In an electromagnetic system, a quantity defined as the ratio of electric dipole moment and magnetic dipole moment has dimension of $\left[M^{P}L^{Q}T^{R}A^{S}\right]$. The value of P and Q are :

**Options :**

A. $-1,0$

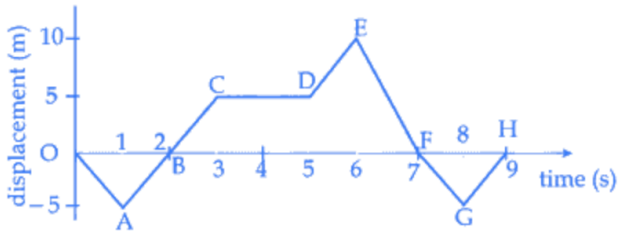
B. $0,-1$

C. $-1,1$

D. $1,-1$

**Question Number : 32 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The displacement x versus time graph is shown below.




(A) The average velocity during 0 to 3 s is 10 m/s

(B) The average velocity during 3 to 5 s is 0 m/s

(C) The instantaneous velocity at t = 2 s is 5 m/s

(D) The average velocity during 5 to 7 s and instantaneous velocity at t = 6.5 s are equal

(E) The average velocity from $t = 0$ to $t = 9$ s is zero

Choose the correct answer from the options given below :

**Options :**

A. (B), (C), (D) only

B. (B), (D), (E) only

C. (A), (D), (E) only

D. (B), (C), (E) only

**Question Number : 33 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider a rectangular sheet of solid material of length $l = 9$ cm and width d $= 4$ cm. The coefficient of linear expansion is $\alpha = 3.1 \times 10^{-5}\ \mathrm{K}^{-1}$ at room temperature and one atmospheric pressure. The mass of sheet $m = 0.1$ kg and the specific heat capacity $C_v = 900\ \mathrm{J\ kg}^{-1}\ \mathrm{K}^{-1}$. If the amount of heat supplied to the material is $8.1 \times 10^2$ J then change in area of the rectangular sheet is :

**Options :**

A. $2.0 \times 10^{-6}\ \mathrm{m}^2$

B. $6.0 \times 10^{-7}\ \mathrm{m}^2$

C. $3.0 \times 10^{-7}\ \mathrm{m}^2$

D. $4.0 \times 10^{-7}\ \mathrm{m}^2$

**Question Number : 34 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

An object is kept at rest at a distance of $3R$ above the earth's surface where $R$ is earth's radius. The minimum speed with which it must be projected so that it does not return to earth is : (Assume M $=$ mass of earth, G $=$ Universal gravitational constant)

**Options :**

A. $\sqrt{\frac{3GM}{R}}$

B. $\sqrt{\frac{2GM}{R}}$

C. $\sqrt{\frac{GM}{2R}}$

D. $\sqrt{\frac{GM}{R}}$

**Question Number : 35 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A cylindrical rod of length 1 m and radius 4 cm is mounted vertically. It is subjected to a shear force of $10^5$ N at the top. Considering infinitesimally small displacement in the upper edge, the angular displacement $\theta$ of the rod axis from its original position would be : (shear moduli, $G = 10^{10}$ N/m$^2$ )

**Options :**

A. $1/160\pi$

B. $1/2\pi$

C. $1/4\pi$

D. $1/40\pi$

**Question Number : 36 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

For the determination of refractive index of glass slab, a travelling microscope is used whose main scale contains 300 equal divisions equals to 15 cm . The vernier scale attached to the microscope has 25 divisions equals to 24 divisions of main scale. The least count (LC) of the travelling microscope is (in cm ) :

**Options :**

A. 0.002

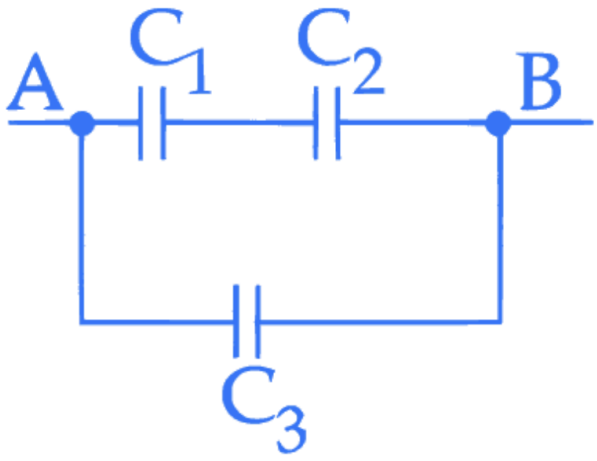
B. 0.0025

C. 0.0005

D. 0.001

**Question Number : 37 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Three parallel plate capacitors $C_1, C_2$ and $C_3$ each of capacitance $5\mu$ F are connected as shown in figure. The effective capacitance between points $A$ and $B$, when the space between the parallel plates of $C_1$ capacitor is filled with a dielectric medium having dielectric constant of 4, is :




**Options :**

A. $9\mu$ F

B. $30\mu$ F

C. $7.5\mu$ F

D. $22.5\mu$ F

**Question Number : 38 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A block of mass 25 kg is pulled along a horizontal surface by a force at an angle 45° with the horizontal. The friction coefficient between the block and the surface is 0.25 . The block travels at a uniform velocity. The work done by the applied force during a displacement of 5 m of the block is :

**Options :**

A. 490 J

B. 970 J

C. 735 J

D. 245 J

**Question Number : 39 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

There are two vessels filled with an ideal gas where volume of one is double the volume of other. The large vessel contains the gas at 8 kPa at 1000 K while the smaller vessel contains the gas at 7 kPa at 500 K . If the vessels are connected to each other by a thin tube allowing the gas to flow and the temperature of both vessels is maintained at 600 K , at steady state the pressure in the vessels will be (in kPa ).

**Options :**

A. 24

B. 4.4

C. 18

D. 6

**Question Number : 40 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Two polarisers $P_1$ and $P_2$ are placed in such a way that the intensity of the transmitted light will be zero. A third polariser $P_3$ is inserted in between $P_1$ and $P_2$, at particular angle between $P_2$ and $P_3$. The transmitted intensity of the light passing the through all three polarisers is maximum. The angle between the polarisers $P_2$ and $P_3$ is :

**Options :**

A. $\pi/6$

B. $\frac{\pi}{3}$

C. $\frac{\pi}{4}$

D. $\pi/8$

**Question Number : 41 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match List - I with List - II.

| | **List - I** | | **List - II** |
|---|---|---|---|
| (A) | Isobaric | (I) | $\Delta Q = \Delta W$ |
| (B) | Isochoric | (II) | $\Delta Q = \Delta U$ |
| (C) | Adiabatic | (III) | $\Delta Q =$ zero |
| (D) | Isothermal | (IV) | $\Delta Q = \Delta U + P\Delta V$ |

$\Delta Q =$ Heat supplied

$\Delta W =$ Work done by the system

$\Delta U =$ Change in internal energy

$P =$ Pressure of the system

$\Delta V =$ Change in volume of the system

Choose the correct answer from the options given below :

**Options :**

A. (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

B. (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

C. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

D. (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

**Question Number : 42 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

There are ' $n$ ' number of identical electric bulbs, each is designed to draw a power $p$ independently from the mains supply. They are now joined in series across the mains supply. The total power drawn by the combination is :

**Options :**

A. np

B. p

C. $\frac{p}{n^2}$

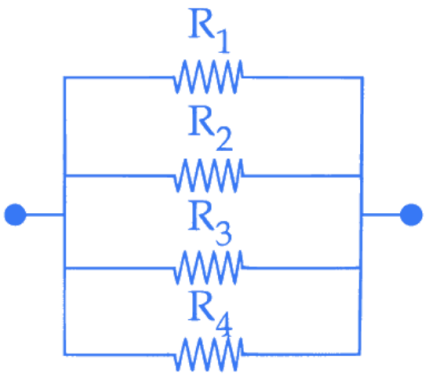
D. $\frac{p}{n}$

**Question Number : 43 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

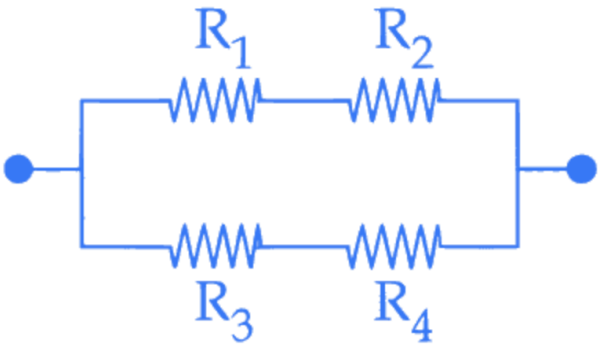
From the combination of resistors with resistances values $R_1 = R_2 = R_3 = 5\Omega$ and $R_4 = 10\Omega$, which of the following combination is the best circuit to get an equivalent resistance of $6\Omega$ ?

**Options :**

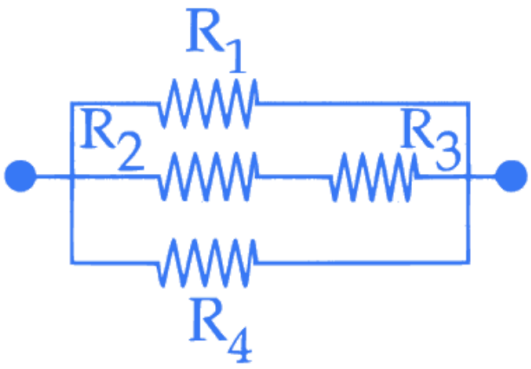
A.



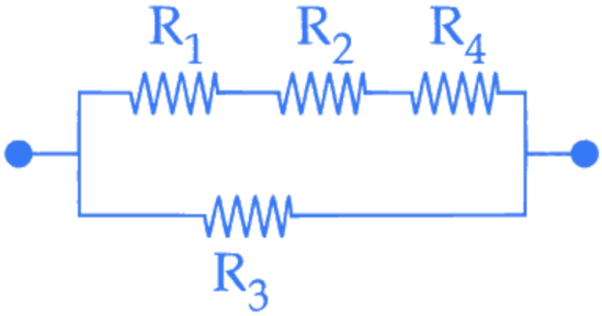

B.




C.




D.



**Question Number : 44 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider a n-type semiconductor in which $n_e$ and $n_h$ are number of electrons and holes, respectively.

(A) Holes are minority carriers

(B) The dopant is a pentavalent atom

(C) $n_e n_h \neq n_i^2$

(where $n_i$ is number of electrons or holes in semiconductor when it is intrinsic form)

(D) $n_e n_h \geqslant n_i^2$

(E) The holes are not generated due to the donors

Choose the correct answer from the options given below :

**Options :**

A. (A), (B), (E) only

B. (A), (C), (E) only

C. (A), (B), (C) only

D. (A), (C), (D) only

**Question Number : 45 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements :

Statement (I) : The dimensions of Planck's constant and angular momentum are same.

Statement (II) : In Bohr's model electron revolve around the nucleus only in those orbits for which angular momentum is integral multiple of Planck's constant.

In the light of the above statements, choose the most appropriate answer from the options given below :

**Options :**

A. Both Statement I and Statement II are correct

B. Statement I is correct but Statement II is incorrect

C. Both Statement I and Statement II are incorrect

D. Statement I is incorrect but Statement II is correct

## Physics Section B

| | |
|---|---|
| **Section Number :** | 4 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |

| | |
|---|---|
| **Number of Questions to be attempted :** | 5 |
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 46 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

A solid sphere with uniform density and radius $R$ is rotating initially with constant angular velocity $(\omega_1)$ about its diameter. After some time during the rotation its starts loosing mass at a uniform rate, with no change in its shape. The angular velocity of the sphere when its radius become R/2 is $x\omega_1$. The value of $x$ is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 47 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

An inductor of self inductance 1 H is connected in series with a resistor of $100\pi$ ohm and an ac supply of $100\pi$ volt, 50 Hz . Maximum current flowing in the circuit is _____ A.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 48 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

If an optical medium possesses a relative permeability of $\frac{10}{\pi}$ and relative permittivity of $\frac{1}{0.0885}$, then the velocity of light is greater in vacuum than that in this medium by _____ times.

$$\left(\mu_0 = 4\pi \times 10^{-7}\text{H/m}, \epsilon_0 = 8.85 \times 10^{-12}\text{ F/m}, \text{c} = 3 \times 10^8\text{ m/s}\right)$$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 49 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

A particle of charge $1.6\mu$C and mass $16\mu$ g is present in a strong magnetic field of 6.28 T . The particle is then fired perpendicular to magnetic field. The time required for the particle to return to original location for the first time is _____ s. $(\pi = 3.14)$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 50 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

In a Young's double slit experiment, two slits are located 1.5 mm apart. The distance of screen from slits is 2 m and the wavelength of the source is 400 nm . If the 20 maxima of the double slit pattern are contained within the central maximum of the single slit diffraction pattern, then the width of each slit is $x \times 10^{-3}$ cm, where $x$-value is _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

## Chemistry Section A

| | |
|---|---|
| **Section Number :** | 5 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

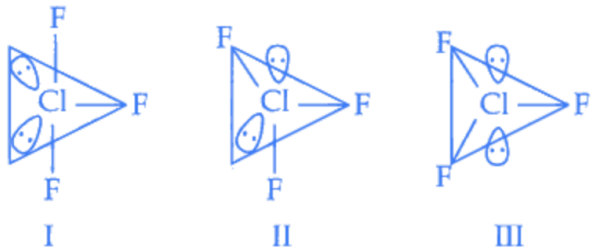
**Question Number : 51 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements:

**Statement (I):** For



, all three possible structures may be drawn as follows.




**Statement (II):** Structure III is most stable, as the orbitals having the lone pairs are axial, where the lp – bp repulsion is minimum.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

**Options :**

A. Both Statement I and Statement II are correct

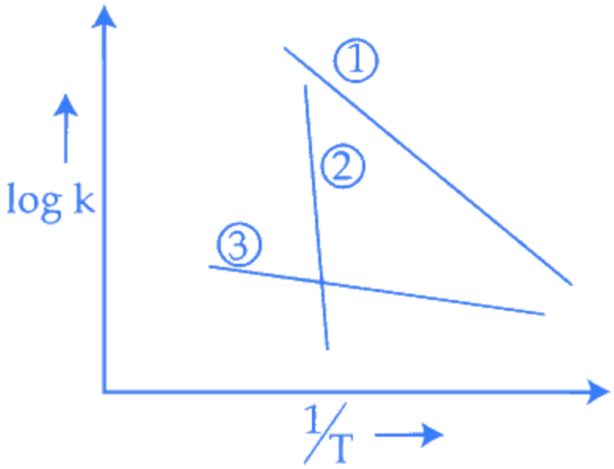
B. Both Statement I and Statement II are incorrect

C. Statement I is correct but Statement II is incorrect

D. Statement I is incorrect but Statement II is correct

**Question Number : 52 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider the following plots of log of rate constant k(log k) vs $\frac{1}{T}$ for three different reactions. The correct order of activation energies of these reactions is :




**Options :**

A. $Ea_2 > Ea_1 > Ea_3$

B. $Ea_1 > Ea_3 > Ea_2$

C. $Ea_3 > Ea_2 > Ea_1$

D. $Ea_1 > Ea_2 > Ea_3$

**Question Number : 53 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match List - I with List - II

| | **List - I (Separation of)** | | **List - II (Separation Technique)** |
|---|---|---|---|
| (A) | Aniline from aniline-water mixture | (I) | Simple distillation |
| (B) | Glycerol from spent-lye in soap industry | (II) | Fractional distillation |
| (C) | Different fractions of crude oil in petroleum industry | (III) | Distillation at reduced pressure |
| (D) | Chloroform-Aniline mixture | (IV) | Steam distillation |

Choose the correct answer from the options given below :

**Options :**

A. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

B. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

C. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

D. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

**Question Number : 54 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements :

Statement (I) : The first ionisation enthalpy of group 14 elements is higher than the corresponding elements of group 13.

Statement (II) : Melting points and boiling points of group 13 elements are in general much higher than those of corresponding elements of group 14.

In the light of the above statements, choose the most appropriate answer from the options given below :

**Options :**

A. Statement I is correct but Statement II is incorrect

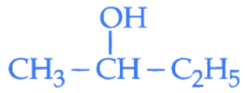
B. Both Statement I and Statement II are incorrect

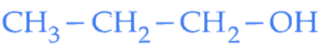
C. Statement I is incorrect but Statement II is correct

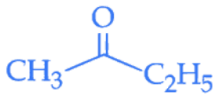
D. Both Statement I and Statement II are correct

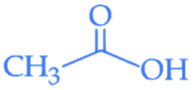
**Question Number : 55 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

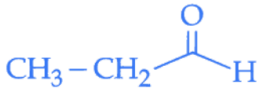
Which among the following compounds give yellow solid when reacted with NaOI/NaOH ?

(A) 

(B) 

(C) 

(D) 

(E) 

Choose the correct answer from the options given below :

**Options :**

A. (A) and (C) Only

B. (C) and (D) Only

C. (B), (C) and (E) Only

D. (A), (C) and (D) Only

**Question Number : 56 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider the ground state of chromium atom ($Z = 24$). How many electrons are with Azimuthal quantum number $l = 1$ and $l = 2$ respectively ?

**Options :**

A. 16 and 5

B. 12 and 5

C. 12 and 4

D. 16 and 4

**Question Number : 57 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements:

Statement (I) : Molal depression constant $K_f$ is given by $\frac{M_1 RT_f}{\Delta S_{fus}}$, where symbols have their usual meaning.

Statement (II) : $K_f$ for benzene is less than the $K_f$ for water.

In the light of the above statements, choose the most appropriate answer from the options given below :

**Options :**

A. Statement I is correct but Statement II is incorrect

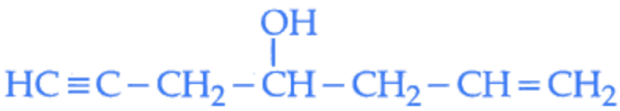
B. Both Statement I and Statement II are incorrect

C. Statement I is incorrect but Statement II is correct

D. Both Statement I and Statement II are correct

**Question Number : 58 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The IUPAC name of the following compound is:



**Options :**

A. 4-Hydroxyhept-6-en-1-yne

B. Hept-1-en-6-yn-4-ol

C. Hept-6-en-1-yn-4-ol

D. 4-Hydroxyhept-1-en-6-yne

**Question Number : 59 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Half life of zero order reaction A $\rightarrow$ product is 1 hour, when initial concentration of reactant is 2.0 mol $L^{-1}$. The time required to decrease concentration of A from 0.50 to 0.25 mol $L^{-1}$ is :

**Options :**

A. 0.5 hour

B. 15 min

C. 60 min

D. 4 hour

**Question Number : 60 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The incorrect relationship in the following pairs in relation to ionisation enthalpies is :

**Options :**

A. $Mn^{2+} < Fe^{2+}$

B. $Mn^{+} < Mn^{2+}$

C. $Mn^{+} < Cr^{+}$

D. $Fe^{2+} < Fe^{3+}$

**Question Number : 61 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A toxic compound " A " when reacted with NaCN in aqueous acidic medium yields an edible cooking component and food preservative " B ". " B " is converted to " C " by diborane and can be used as an additive to petrol to reduce emission. "C" upon reaction with oleum at 140°C yields an inhalable anesthetic " D ". Identify " A ", " B ", " C " & " D ", respectively :

**Options :**

A. Methanol; acetic acid; ethanol; diethyl ether

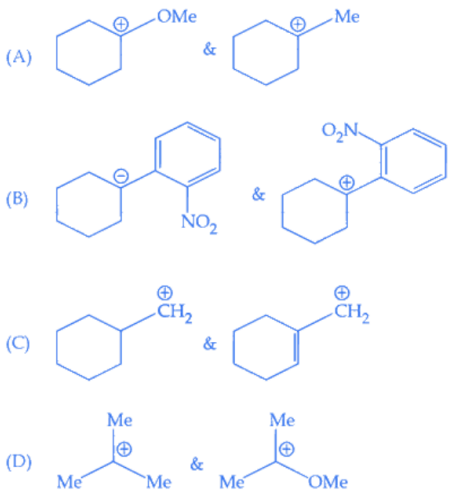
B. Ethanol; acetonitrile; ethylamine; ethylene

C. Methanol; formaldehyde; methyl chloride; chloroform

D. Acetaldehyde; 2-hydroxypropanoic acid; propanoic acid; dipropyl ether

**Question Number : 62 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

In which pairs, the first ion is more stable than the second?




**Options :**

A. (A) & (B) only

B. (B) & (D) only

C. (A) & (C) only

D. (B) & (C) only

**Question Number : 63 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

'$X$' is the number of electrons in $t_{2g}$ orbitals of the most stable complex ion among $\left[Fe(NH_3)_6\right]^{3+}$, $\left[FeCl_6\right]^{3-}$, $\left[Fe(C_2O_4)_3\right]^{3-}$ and $\left[Fe(H_2O)_6\right]^{3+}$. The nature of oxide of vanadium of the type $V_2O_X$ is :

**Options :**

A. Amphoteric

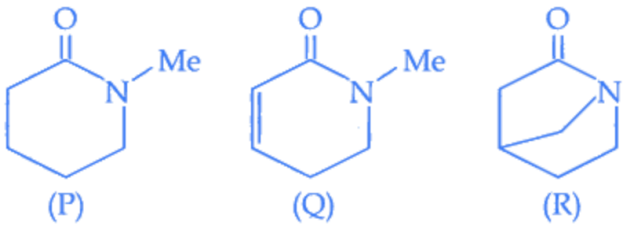
B. Acidic

C. Basic

D. Neutral

**Question Number : 64 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The correct order of basicity for the following molecules is :




**Options :**

A. $R > P > Q$

B. $P > Q > R$

C. $R > Q > P$

D. $Q > P > R$

**Question Number : 65 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider the given data :

(a) $HCl(g) + 10H_2O(l) \rightarrow HCl.10H_2O \Delta H = -69.01 \text{ kJ mol}^{-1}$

(b) $HCl(g) + 40H_2O(l) \rightarrow HCl.40H_2O \Delta H = -72.79 \text{ kJ mol}^{-1}$

Choose the correct statement :

**Options :**

A. The heat of dilution for the $HCl\left(HCl.10H_2O \text{ to } HCl.40H_2O\right)$ is 3.78 kJ $\text{mol}^{-1}$.

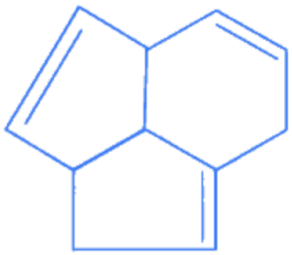
B. Dissolution of gas in water is an endothermic process.

C. The heat of solution depends on the amount of solvent.

D. The heat of formation of HCl solution is represented by both (a) and (b).

**Question Number : 66 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

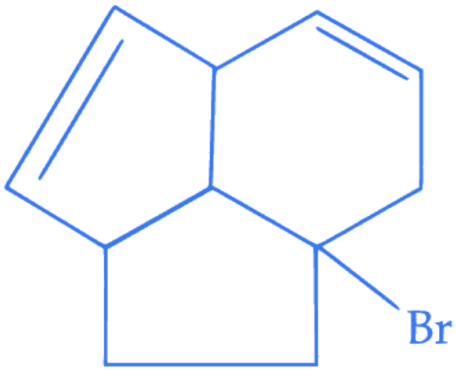
Consider the following molecule(X).

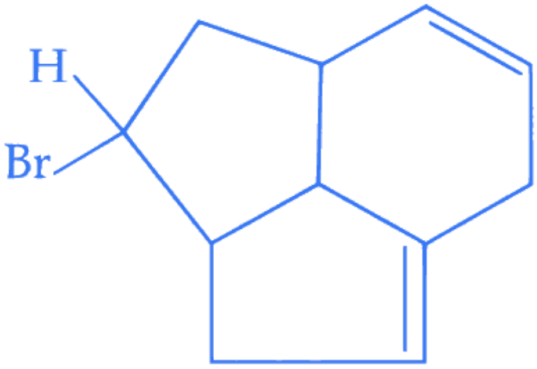
The structure of X is 

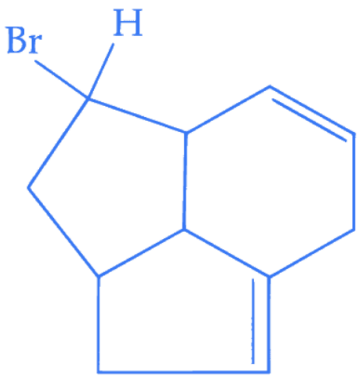
(X)

The major product formed when the given molecule (*X*) is treated with HBr(1eq) is :

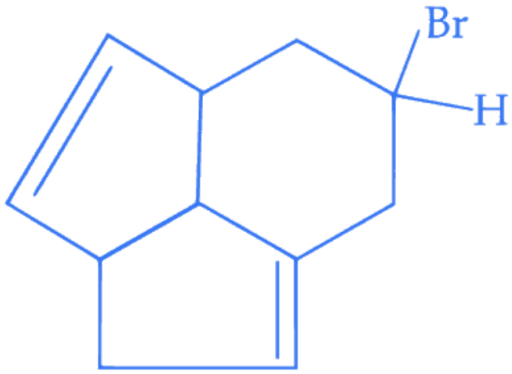
**Options :**

A. 


B. 


C. 

D.




**Question Number : 67 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The correct order of $[FeF_6]^{3-}$, $[CoF_6]^{3-}$, $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$ complex species based on the number of unpaired electrons present is:

**Options :**

A. $[CoF_6]^{3-} > [FeF_6]^{3-} > [Ni(CO)_4] > [Ni(CN)_4]^{2-}$

B. $[FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CN)_4]^{2-} > [Ni(CO)_4]$

C. $[FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CN)_4]^{2-} = [Ni(CO)_4]$

D. $[Ni(CN)_4]^{2-} > [FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CO)_4]$

**Question Number : 68 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements :

**Statement (I) :** Alcohols are formed when alkyl chlorides are treated with aqueous potassium hydroxide by elimination reaction.

**Statement (II) :** In alcoholic potassium hydroxide, alkyl chlorides form alkenes by abstracting the hydrogen from the $\beta$-carbon.

In the light of the above statements, choose the most appropriate answer from the options given below :

**Options :**

A. Both Statement I and Statement II are incorrect

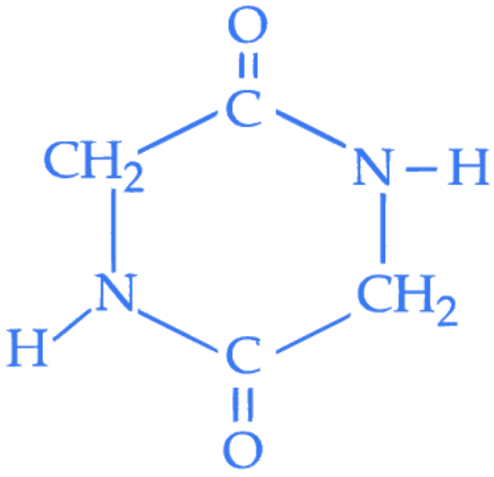
B. Both Statement I and Statement II are correct

C. Statement I is incorrect but Statement II is correct

D. Statement I is correct but Statement II is incorrect

**Question Number : 69 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A dipeptide, " $x$ " on complete hydrolysis gives " $y$ " and " $z$ ". " $y$ " on treatment with aq. $HNO_2$ produces lactic acid. On the other hand " z " on heating gives the following cyclic molecule.



Based on the information given, the dipeptide $X$ is :

**Options :**

A. alanine-alanine

B. valine-glycine

C. valine-leucine

D. alanine-glycine

**Question Number : 70 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The elements of Group 13 with highest and lowest first ionisation enthalpies are respectively :

**Options :**

A. B & Tl

B. Tl & B

C. $B$ & In

D. $B$ & $Ga$

## Chemistry Section B

| | |
|---|---|
| **Section Number :** | 6 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |
| **Number of Questions to be attempted :** | 5 |

| | |
|---|---|
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 71 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Sea water, which can be considered as a 6 molar (6M) solution of NaCl , has a density of 2 g $mL^{-1}$. The concentration of dissolved oxygen $(O_2)$ in sea water is 5.8 ppm . Then the concentration of dissolved oxygen $(O_2)$ in sea water, is $x \times 10^{-4}$ m.

$x =$ _____. (Nearest integer)

Given: Molar mass of NaCl is 58.5 g $mol^{-1}$

Molar mass of $O_2$ is 32 g $mol^{-1}$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 72 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

The amount of calcium oxide produced on heating 150 kg limestone ( 75% pure) is _____ kg. (Nearest integer)

Given: Molar mass (in $gmol^{-1}$ ) of Ca-40, O-16, C-12

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 73 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

A metal complex with a formula $MCl_4 \cdot 3NH_3$ is involved in $sp^3 d^2$ hybridisation. It upon reaction with excess of $AgNO_3$ solution gives ' $x$ ' moles of AgCl . Consider ' $x$ ' is equal to the number of lone pairs of electron present in central atom of $BrF_5$. Then the number of geometrical isomers exhibited by the complex is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 74 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

The molar conductance of an infinitely dilute solution of ammonium chloride was found to be 185 S $cm^2$ $mol^{-1}$ and the ionic conductance of hydroxyl and chloride ions are 170 and 70 S $cm^2$ $mol^{-1}$, respectively. If molar conductance of 0.02 M solution of ammonium hydroxide is 85.5 S $cm^2$ $mol^{-1}$, its degree of dissociation is given by $x \times 10^{-1}$. The value of $x$ is _____ . (Nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 75 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

$x$mg of $Mg(OH)_2$( molar mass $= 58$) is required to be dissolved in 1.0 L of water to produce a pH of 10.0 at 298 K . The value of $x$ is _____ mg. (Nearest integer)

(Given : $Mg(OH)_2$ is assumed to dissociate completely in $H_2O$ ]

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText